London Worm Club Meeting Nov 2026

Date: Wed 4th Nov 2026
Time: 2-5 pm, followed by drinks reception
MRC Laboratory of Medical Sciences, Hammersmith Campus, W12 0HS. Directions

Sponsored by: The Genetics Society

Agenda

14:00 **Barbara Conradt, UCL**
Mitochondrial inheritance and cell fate decisions during asymmetric neuroblast divisions

14:25 **Charles Fieseler (Zimmer lab), University of Vienna**
Neuronal principles underlying the hierarchical organization of animal behavior

14:50 **Lightning talks x 5**
Alan Koh, MRC Laboratory of Medical Sciences
Chee Kiang (Ethan) Ewe, Imperial College,
Rosie Coleman, Flinders University
Davide Pasini, Labbadia Lab, University College London
Pablo López-Jiménez, MRC Laboratory of Medical Sciences

15:15 ***Coffee break***

15:40 **Sean Wallace (Angika Basant lab), KCL**
Using *C. elegans* as a model to explore the functions of PRL phosphatases

16:05 **Om Patange, UCL**
The importance of sulphur metabolism in animal–Actinobacteria interactions

16:30 **Erna King (Blaxter lab), Sanger Institute**
Exploring the worm's evolutionary history with single wild specimens across nematode diversity

16:55 **Closing announcements**

17:00 **Drinks and snacks**

Abstracts

Barbara Conradt, UCL

Mitochondrial inheritance and cell fate decisions during asymmetric neuroblast divisions

We use *C. elegans* Q lineage neuroblasts to study the inheritance of organelles during asymmetric cell divisions and to determine the impact of organelle inheritance on cell fate decisions. Using live super-resolution imaging, we found that mitochondria segregate unequally by density and morphology during QL.p division, which divides asymmetrically to generate a smaller daughter cell that undergoes apoptosis (QL.pp) and a larger daughter cell that divides to give rise to two neurons (PVM and SDQL) (QL.pa). In addition, we found that mitochondrial density in QL.pp negatively correlates with the ability of QL.pp to die. I will give an update on our efforts to determine the mechanisms involved in unequal mitochondrial segregation and to address the question whether mitochondria in QL.pp have anti-apoptotic activity.

Charles Fieseler (Zimmer lab), University of Vienna

Neuronal principles underlying the hierarchical organization of animal behavior

Charles Fieseler*, Itamar Lev*, Ulises Rey*, Lukas Hille, Hannah Brenner, Manuel Zimmer
* Co-first authors

Understanding the neuronal mechanisms underlying behavioral hierarchies is a long standing question in neuroscience. One cross-species observation made in behaving animals is that widely distributed neuronal activity patterns are related to instantaneous behavior. When performing repetitive, isolated actions, neuronal population activity is well modeled via smooth trajectories in lower dimensional subspaces, termed manifolds. However, it is unknown whether manifolds also encode behaviors at different hierarchical levels and across full sequences of behaviors. Using a new pipeline for single-cell resolution whole brain imaging in freely moving *C. elegans*, we discover that the neuronal activity distributed across the entire brain is dominated by a low-dimensional manifold which encodes the entire natural action sequence of the animal. Further, we discover a functional

relationship between the neuronal representation of the action sequence and fine-scale activity in the motor periphery, corresponding to gait patterns. We use a nonlinear gating model, which models the peripheral neurons as selectively encoding faster time-scale motor patterns during specific brain states. In this model, information about major actions is broadcast via a distributed manifold, enabling nesting of fine-scale motor patterns within those major actions. Enabling such hierarchical organization of behavior may be a general principle of neuronal manifold structures.

Sean Wallace (Angika Basant lab), KCL

Using *C. elegans* as a model to explore the functions of PRL phosphatases

Phosphatases are a large group of enzymes that play essential roles in cellular signaling pathways, yet their regulation is relatively poorly understood compared to their kinase counterparts. We study how phosphatases regulate the actin cytoskeleton using Vaccinia virus-infected human cells, and we identified PRL phosphatases as candidate regulators of the actin cytoskeleton through RNAi screens. PRL phosphatases are a subfamily of dual specificity phosphatases comprising 3 genes in mammals. PRL phosphatases are overexpressed in many metastatic cancers, and cell culture studies demonstrated roles in cell proliferation and migration. Single gene knockout mouse models exhibit mild phenotypes, while double knockouts are lethal, suggesting redundancy. In vitro studies showed that PRL enzymatic activity is weak compared to other phosphatases, and additional non-enzymatic functions have been proposed including magnesium homeostasis and allosteric regulation of Src kinase activity. Overall the functions of PRL phosphatases in vivo remain unclear.
In parallel with our human cell culture model, we recently started exploring the function of the single *C. elegans* PRL phosphatase gene, ptp-4A2. To characterize ptp-4A2 expression, we generated an endogenous knock-in fluorescent reporter strain. ptp-4A2 is expressed broadly in a number of tissues, including the reproductive system, the nervous system and the digestive system. Within the reproductive system ptp-4A2 is expressed in several somatic cell types, including the distal tip cell (DTC) and the gonadal sheath cells, but not in germ cells. The DTC and gonadal sheath cells both regulate germline homeostasis in a variety of ways, and we are currently exploring the role of ptp-4A2 in these processes.
The distal tip cell (DTC) is a large elaborately-shaped cell with multiple membrane extensions. The DTC forms a stem cell niche by contacting germ stem cells (GSCs) and maintaining their undifferentiated state in the progenitor zone of the distal gonad. Elaboration of DTC morphology is therefore essential for maintaining the GSC niche. We generated a ptp-4A2 deletion strain, and found that DTC morphology is perturbed: membrane processes do not extend as far along the distal gonad. This is accompanied by a reduction in the size of the progenitor zone and a reduced brood size.
Gonadal sheath cells surround the gonad and regulate germline homeostasis through phagocytic engulfment of apoptotic germ cells. We observe PTP-4A2 protein localization outlining presumptive apoptotic germ cells in a manner reminiscent of sheath cell membrane envelopment of apoptotic corpses.
We plan to explore the roles of PTP-4A2 in regulating both DTC morphology and sheath cell engulfment of apoptotic germ cells, with a particular interest in cytoskeletal and membrane dynamics. We will use live imaging and genetic epistasis experiments to characterize

ptp-4A2 phenotypes in detail. We have generated a panel of point mutations in the ptp-4A2 gene corresponding to functional mutations defined in human PRL proteins, and we will use these for structure-function studies to elucidate the molecular mechanisms through which PTP-4A2 acts. Insights from these experiments will complement our ongoing studies of human PRL proteins, whose molecular and cellular functions have remained enigmatic.

Om Patange, UCL

The importance of sulphur metabolism in animal–Actinobacteria interactions

Actinobacteria can be pathogenic to animals. For example, *Mycobacterium tuberculosis* causes tuberculosis in humans, while *Microbacterium* sp. JUb115, a bacterium isolated from the natural ecology of *C. elegans*, causes nematodes to grow slowly. Surprisingly, sulphur metabolites such as hydrogen sulphide (H_2S), and the regulators that control their levels, are key determinants of the outcome of these animal–microbe interactions. Here, I will explore the role of the transsulfuration pathway in these interactions, building on discoveries from unbiased forward genetics. We find that genetically reducing H_2S production improves the survival of *C. elegans* infected with *Mycobacterium marinum*. However, the metabolic requirements for growth on JUb115 appear strikingly different: mutations that reduce H_2S production most strongly instead cause larval arrest. A genetic screen for mutations that suppress this arrest reveals unexpected complexity in the regulation of sulphur metabolism. This work helps answer the fundamental question: how do animals regulate their metabolism when different pathogens favour different levels of the same metabolites?

Erna King (Mark Blaxter lab), Sanger Institute

Exploring the worm's evolutionary history with single wild specimens across nematode diversity

Erna King, Lewis Stevens, Witek Morek, Oleksandr Holovachov*, Chris Laumer** & Mark Blaxter
Tree of Life, Wellcome Sanger Institute, Cambridge, United Kingdom
*Natural History Museum, Sweden
**Natural History Museum London

When exploring model organisms in the 1960s to study development, sex determination and mutation in a system "amenable to genetic analysis" Sydney Brenner was swayed largely by how easily inbred strains of nematodes could be cultured and manipulated. Huge strides have been made since "the worm", *Caenorhabditis elegans*, became the first multicellular organism for which a genome was sequenced, but this success has not been replicated across the diversity of the phylum. Sequencing nematodes beyond those that are culturable or large parasites has been hampered because single individuals yield too little DNA for existing low-input sequencing techniques whilst pooled specimens are too heterozygous for genome assembly. This has led to an imbalance in genomic representation across the phylum. We have used our novel picogram input protocol to overcome this input limitation, generating high quality genomes and transcriptomes for 37 wild nematode species from single individuals. These genomes span 3 classes, 14 orders and 23 families, providing the first genome for 3 orders and 12 families. Eight dominant species were also collected in

sufficient abundance to pool individuals for ultra-low input Hi-C scaffolding, yielding the first chromosomal level assemblies for the class Enoplea, and for 5 orders and 7 families. Together with published chromosome-level genomes, these genomes support the conservation of the ancestral linkage groups first described for Rhabditida, termed Nigon elements, deeper into Chromadorea, where most lineages have undergone relatively few rearrangements. Beyond Chromadorea, chromosomes of the free-living Dorylaimea have undergone significant rearrangement, while conserving Nigon partitioning. The first chromosomal genome for the basal class, Enoplea, revealed extensive rearrangement and no signal of Nigon elements, suggesting that the Nigon model is not supported across the phylum. Together, these wild genomes promise insight into the evolutionary history of *C. elegans* and friends, expanding the relevance of the knowledge gleaned from nematodes studied since Sydney Brenner chose the worm over sixty years ago.

Lightning talk abstracts

Alan Koh, MRC Laboratory of Medical Sciences
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LITE-1: A multimodal sensory receptor for light and chemical stimuli in *C. elegans*

The nematode *Caenorhabditis elegans* can detect and respond to a wide range of environmental stimuli through specialised sensory receptors. Among these is the gustatory receptor homolog LITE-1, which plays an essential role in mediating aversive responses to blue and ultraviolet light. In our recently published work, we demonstrated that LITE-1 also functions as a chemoreceptor, mediating avoidance of high concentrations of the odorants diacetyl and 2,3-pentanedione. This unexpected finding establishes LITE-1 as a multimodal receptor involved in both photosensation and chemosensation. However, whether these two functions share the same downstream signalling components remains unknown. We therefore examined whether components required for LITE-1-dependent photosensation are also involved in diacetyl avoidance. Although photosensation involves a well-characterised cyclic guanosine monophosphate (cGMP) signalling pathway, our behavioural analyses indicated that most mutants lacking individual components of this pathway retained their ability to avoid high concentrations of diacetyl. This suggests that LITE-1-mediated chemosensation may involve signalling mechanisms distinct from those required for photosensation. Together, our findings reveal that LITE-1 can mediate responses to both light and chemical stimuli, potentially through different downstream signalling pathways.

Chee Kiang (Ethan) Ewe, Imperial College
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Oxygen-sensing circuit shapes germline thermotolerance

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Thermal pollution, whether local or driven by global warming, threatens biodiversity in part through its detrimental effects on reproduction. Non-coding small RNAs (sRNAs) are crucial for maintaining germline developmental robustness under heat stress. Recently, we found that, in RNAi mutants, an oxygen-sensing neural circuit, modulated by neuropeptide signaling, antagonizes germline maintenance, likely reflecting the nematode's innate association of reduced oxygen levels with food availability and reproductive permissive environments. Hence, our findings raise the possibility that sensory perception, independent of direct environmental change, modulates germline integrity, highlighting a novel mechanism by which neural circuits integrate environmental information to safeguard reproductive fitness in fluctuating environments.

Rosie Coleman, Flinders University
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Tissue-specific proteostasis responses in *C. elegans* suggest a protective role for protein aggregation in ageing outcomes

Flinders Health and Medical Research Institute, Flinders University, South Australia
Chew Worm Lab (Currently visiting the Schafer lab)

Protein aggregation is a well-established hallmark of proteostasis dysfunction. To investigate the tissue-specific effects of proteostasis challenge on organismal ageing, we employed *C. elegans* expressing either stable wild-type (WT) or aggregation-prone double-mutant (DM) firefly luciferase (FLuc), fused to EGFP, in muscle or neurons. Longitudinal confocal imaging revealed distinct aggregation patterns between tissues. Muscle- FLucDM formed aggregates starting from young adulthood and persisting to old age, while muscle- FLucWT worms remained diffuse at all ages. In contrast, both FLucWT and FLucDM in neurons showed aggregates throughout their lifespan, with neural- FLucDM showing an overall higher aggregate burden. Unexpectedly, aggregate formation was not associated with poorer ageing outcomes. Lifespan and healthspan assays revealed that despite the presence of aggregates, muscle-specific FLucDM worms were significantly longer-lived compared with muscle-FLucWT and maintained youthful locomotor function for longer. In contrast, neuronal FLucDM animals showed no lifespan advantage over neuronal FLucWT controls. However, they exhibited improved youthful locomotor healthspan, suggesting that some beneficial effects of aggregation can be uncoupled from longevity. Together, these observations indicate that the relationship between protein aggregation and ageing is tissue-specific. This suggests that aggregate burden alone does not predict organismal ageing outcomes, and in some contexts, may reflect a protective mechanism.

Davide Pasini (Labbadia Lab), University College London
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Does retuning the heat shock response promote fitness?

Ageing is a complex and multifactorial process characterised by the loss of protein homeostasis, a cellular state in which all proteins are working optimally. The maintenance of a healthy proteome is a task performed by quality control cellular pathways, among which the Heat Shock Response (HSR) plays an essential role to correct and remove damaged proteins from cells. Research that used *Caenorhabditis elegans* as model animal has challenged the conventional supposition that ageing is a gradual process caused by the random accumulation of damage. Instead of the progressive deterioration, these studies have shown that the loss of protein integrity occurs in a programmed way through repression of HSR by signals from the germline, just before animals commence reproduction. In particular, the HSR repression has been found to coincide with the onset of yolk protein (vitellogenin) mass production by the intestinal cells to support post-embryonic development and larval robustness. Based on this, we hypothesise that the HSR is actively retuned in adult *C. elegans* to maximise the rapid production of yolk, thereby highlighting a trade-off between reproductive fitness and long-term proteome health. Currently, we are investigating whether the HSR repression is conserved in other hermaphroditic Caenorhabditis species as well as their gonochoristic (male/female) sibling species.

Pablo López-Jiménez, MRC Laboratory of Medical Sciences

Resolving the molecular basis of meiotic chromosome structure using biophysical and in vivo approaches

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Cohesin complexes play crucial roles in both mitotic and meiotic chromosomes, including sister chromatid cohesion (SCC), DNA damage repair, and 3D genome organisation. During meiosis, chromosomes undergo significant structural changes that are necessary for the formation of haploid gametes from diploid germ cells. These changes include the formation of axial elements that promote pairing and recombination between homologous chromosomes, which, together with SCC, ensure correct chromosome segregation into the gametes. These key meiotic events require cohesin complexes containing meiosis-specific kleisins that display functional specialisation in mammals and *Caenorhabditis elegans*: REC-8REC8 complexes ensure SCC and contribute to axis assembly, while COH-3/4RAD21L complexes promote axis assembly and the organisation of chromosomes into higher-order structures. However, how the modular design of meiotic cohesin gives rise to complexes with distinct functions remains unclear. To address this, we combine single-molecule imaging with in vivo experiments in *C. elegans* to determine whether REC-8 and COH-3/4 complexes display distinct molecular activities on DNA. Our findings support a

model in which the identity of the kleisin subunit defines functionally distinct cohesin complexes that coexist on chromosomes, mediating SCC and DNA looping.